COMPUTER NETWORK LAB PROJECT

For

HOTEL NETWORK CONFIGURATION

Version 1.0

Prepared by

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Submitted to

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Introduction

This project entails the design and configuration of a comprehensive network infrastructure tailored for a hotel environment, utilizing Cisco Packet Tracer. The network setup encompasses various elements including serial router configurations, password security measures and DHCP implementation to streamline IP address management.

Objective

The primary aim of this project is to develop a robust and efficient network infrastructure customized to meet the specific demands of a hotel setting. This involves the creation of segmented VLANs to manage traffic effectively, configuring routers to enable inter-VLAN communication, implementing security measures such as password protection, deploying a web server to facilitate guest services, and integrating DHCP services for simplified IP address allocation.

Description of System:

Admin Office: Located on the first floor, Router 1 serves as the gateway for this area. It is responsible for managing network traffic and providing internet access to administrative devices.

Reception Area: Connected to the Admin Office, this area is the focal point for guest interactions. Devices in the reception area connect to the Admin Office router for network services.

Marketing Department (First Floor): Router 2 is connected to this department, applying STICK VLAN segmentation to isolate its network traffic. It ensures efficient communication within the marketing department while maintaining security.

IT Department (First Floor): Also connected to Router 7, the IT department has its STICK VLAN to separate its network operations. This setup allows the IT team to manage their resources independently.

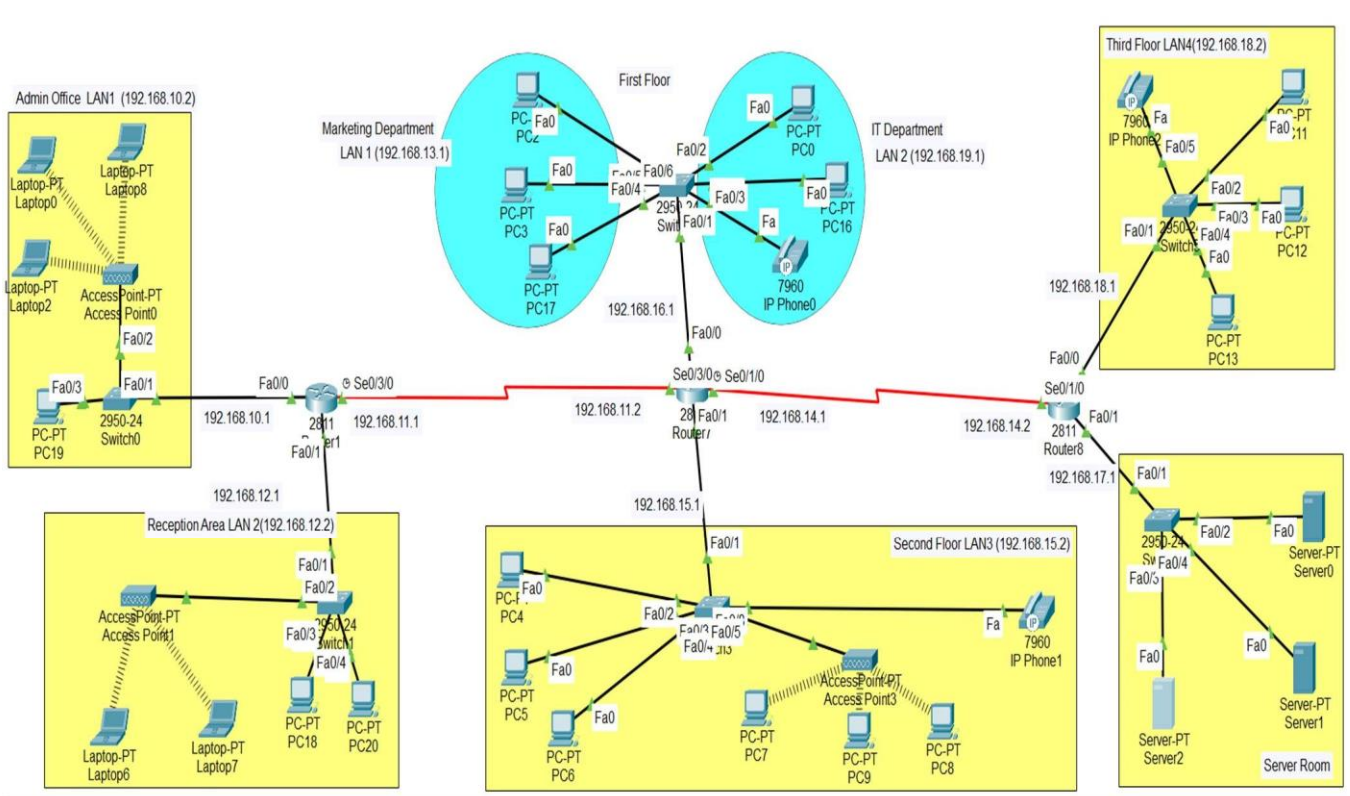
Second Floor and Third Floor: These floors follow a similar network setup as the first floor, with routers connecting to the central router (Router 2) for internet access and STICK-VLAN communication.

Server Room: Located on the third floor, Router 3 connects this critical area to the central router. It ensures that servers hosting essential hotel services are accessible to all relevant departments while maintaining security measures.

Network Topology

The network topology comprises the following components:

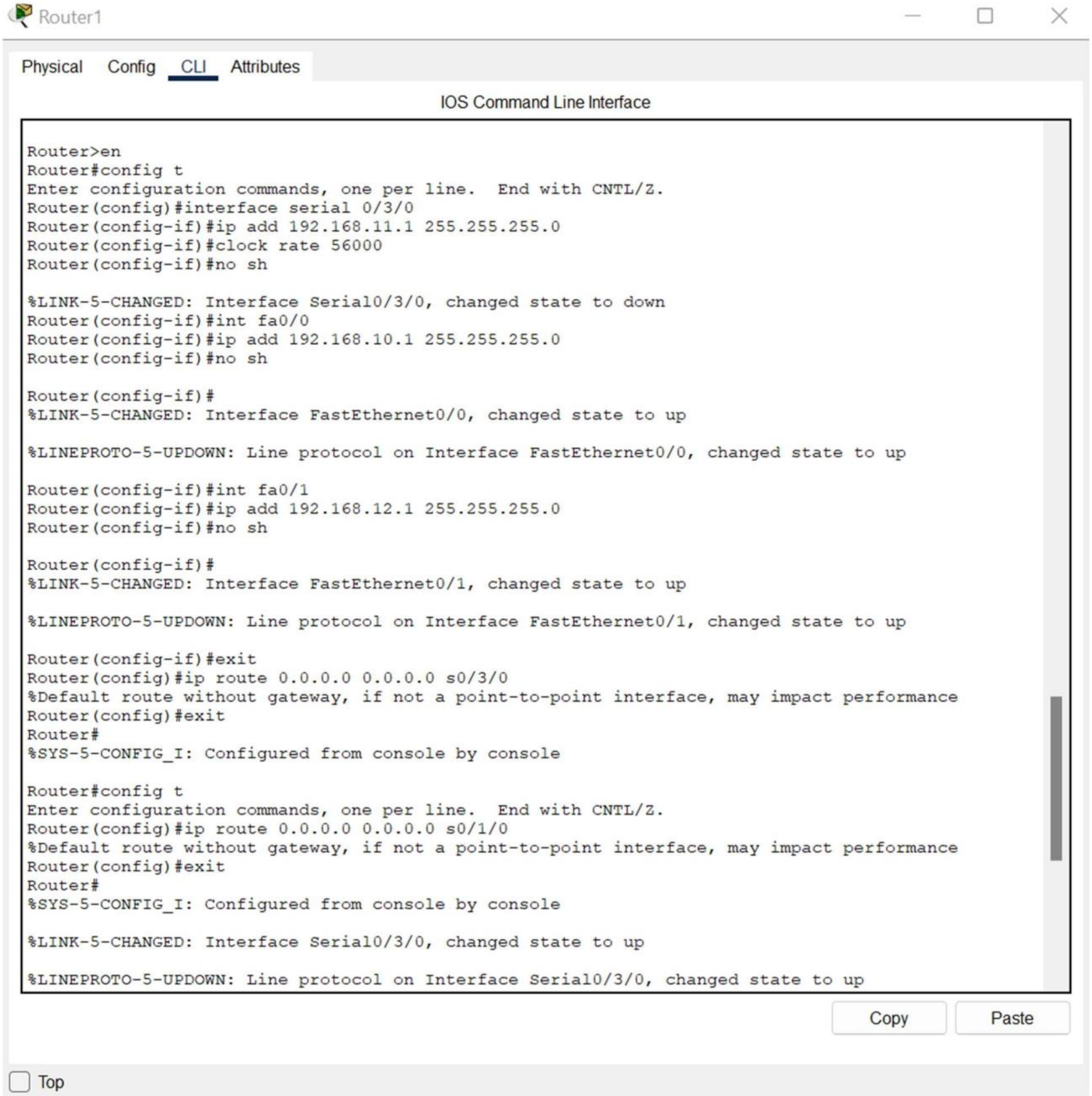
1. Routers: Facilitate interconnection between VLANs and provide gateway access to the internet.
2. Switches: Serve to connect devices within each VLAN segment.
3. PCs: Representing guest devices including laptops, smartphones.
4. Web Server: Hosts essential services such as guest portal, room service requests, etc.
5. DHCP Server: Facilitates automatic IP address assignment to network devices.



Configuration Details

1. Serial Router Configuration

Router 1


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.1 255.255.255.0
Router(config-if)#clock rate 56000
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.10.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.12.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
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```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 s0/3/0
                                ^
% Invalid input detected at '^' marker.

Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
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Router 2

Router7

Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Router(config-if)#interfac
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed stat
Router(config-if)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#ip add 192.168.14.1 255.255.255.0
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.16.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.15.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

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```
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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Router 3

Router8

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface s0/1/0
Router(config-if)#ip add 192.168.14.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

Router(config-if)#e
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
xi
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.18.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.17.1 255.255.255.0
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip add 192.168.17.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

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```
Router#en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

2. Stick VLAN Implementation
   Switch

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name it
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
10   marketing                        active
20   it                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/4-6
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br
```

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
10   marketing                        active
20   it                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

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Router 2

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#no sh
Router(config-if)#int f0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.13.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int f0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.19.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

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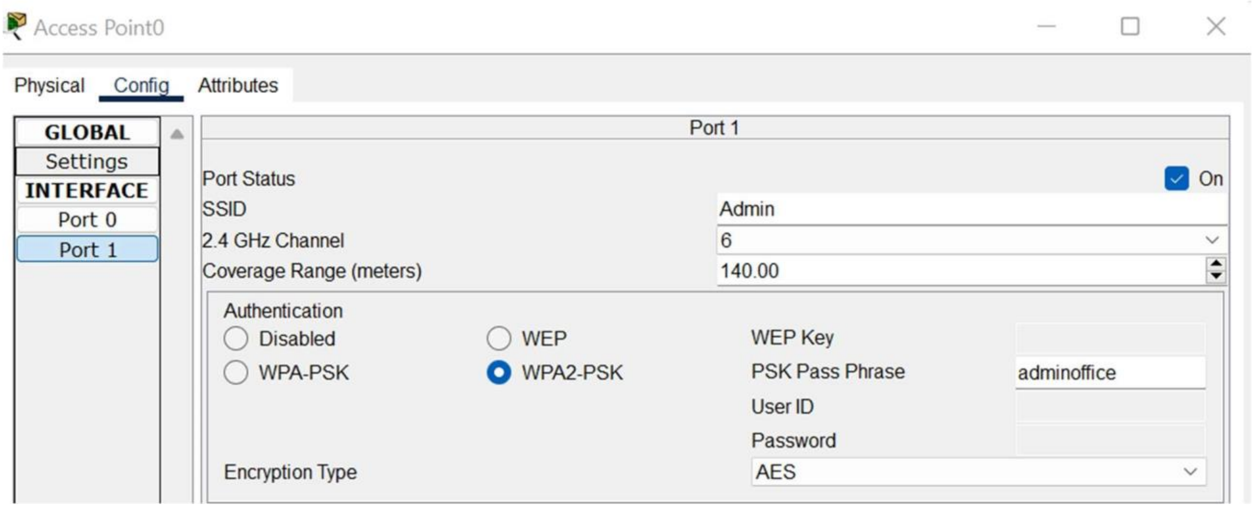
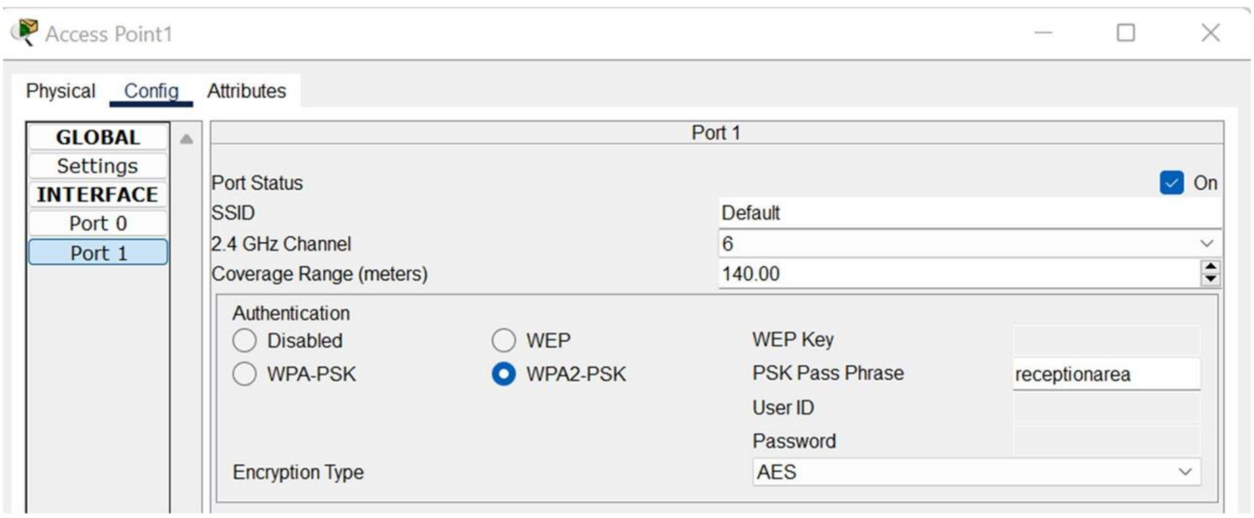
3. Password Setting

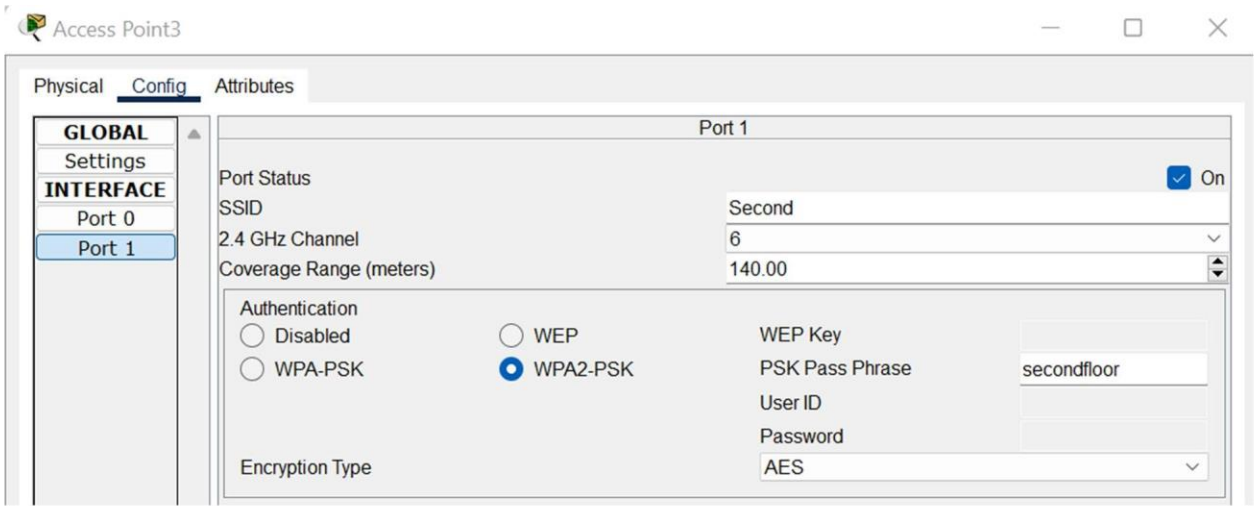
Admin Office



Reception Area



Second Floor



4. DHCP Implementation

Router 1

```
Router#en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool lan1
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#network 192.168.10.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#
```

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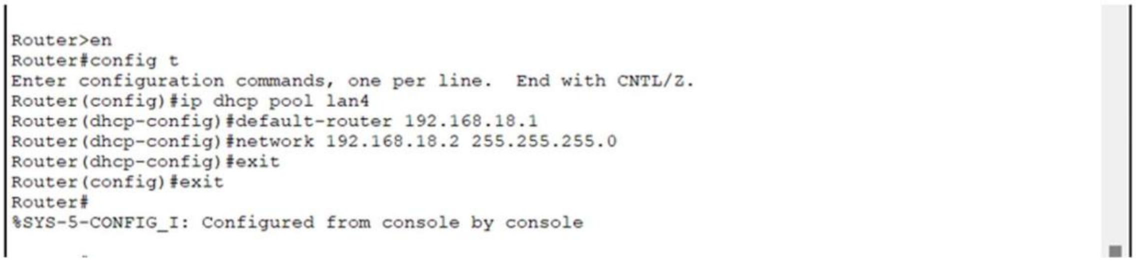
Top

Router 2

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool lan2
Router(dhcp-config)#default-router 192.168.16.1
Router(dhcp-config)#network 192.168.16.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp pool lan3
Router(dhcp-config)#default-router 192.168.15.1
Router(dhcp-config)#network 192.168.15.2  255.255.255.0
Router(dhcp-config)#exit
```

Router 3



```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip dhcp pool lan4
Router(dhcp-config)#default-router 192.168.18.1
Router(dhcp-config)#network 192.168.18.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Conclusion

By implementing this network infrastructure, the hotel ensures efficient communication, robust security, and seamless access to essential services for guests and staff. The segmented STICK VLAN, DHCP, Passwords and routing configuration, facilitate smooth operations across different departments while maintaining network integrity and security.